

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** Thermal Physics **L-T-P:** 3-1-0 **Credits:** 4

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

Thermodynamic systems, The First law, Ideal Gas, The Second law, Carnot's engine, The Entropy, Thermodynamic potentials, Open systems, Phase transition. Nernst's theorem. (10)

Blackbody radiation, equation of state, density of states, Planck's law. (2)

Random walk, Elementary kinetic theory, Boltzmann equation. Maxwell- Boltzmann distribution, specific heat capacity, Transport phenomena and Diffusion. (16)

Elements of Information theory, Shanon entropy, Information and thermodynamics, Data compression, Quantum information. (8) Introductory ideas of statistical mechanics: ensembles, partition functions and their relationship with thermodynamics. (6)

Reference Books:

1. Thermodynamics: E. Fermi
2. Concepts in thermal physics: S. J. Blundell and K. M. Blundell
3. An introduction to thermal Physics : D.V. Schroeder
5. Thermal Physics: C. Kittel

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Thermodynamic systems, The First law, Ideal Gas, The Second law, Carnot's engine, The Entropy, Thermodynamic potentials, Open systems, Phase transition. Nernst's theorem.	10
2.	Blackbody radiation, equation of state, density of states, Planck's law.	2
3.	Random walk, Elementary kinetic theory, Boltzmann equation. Maxwell- Boltzmann distribution, specific heat capacity, Transport phenomena and Diffusion.	16
4.	Elements of Information theory, Shanon entropy, Information and thermodynamics, Data compression, Quantum information.	8
5.	Introductory ideas of statistical mechanics: ensembles, partition functions and their relationship with thermodynamics.	6
Total number of hours		42